EXPERIMENT INDEX

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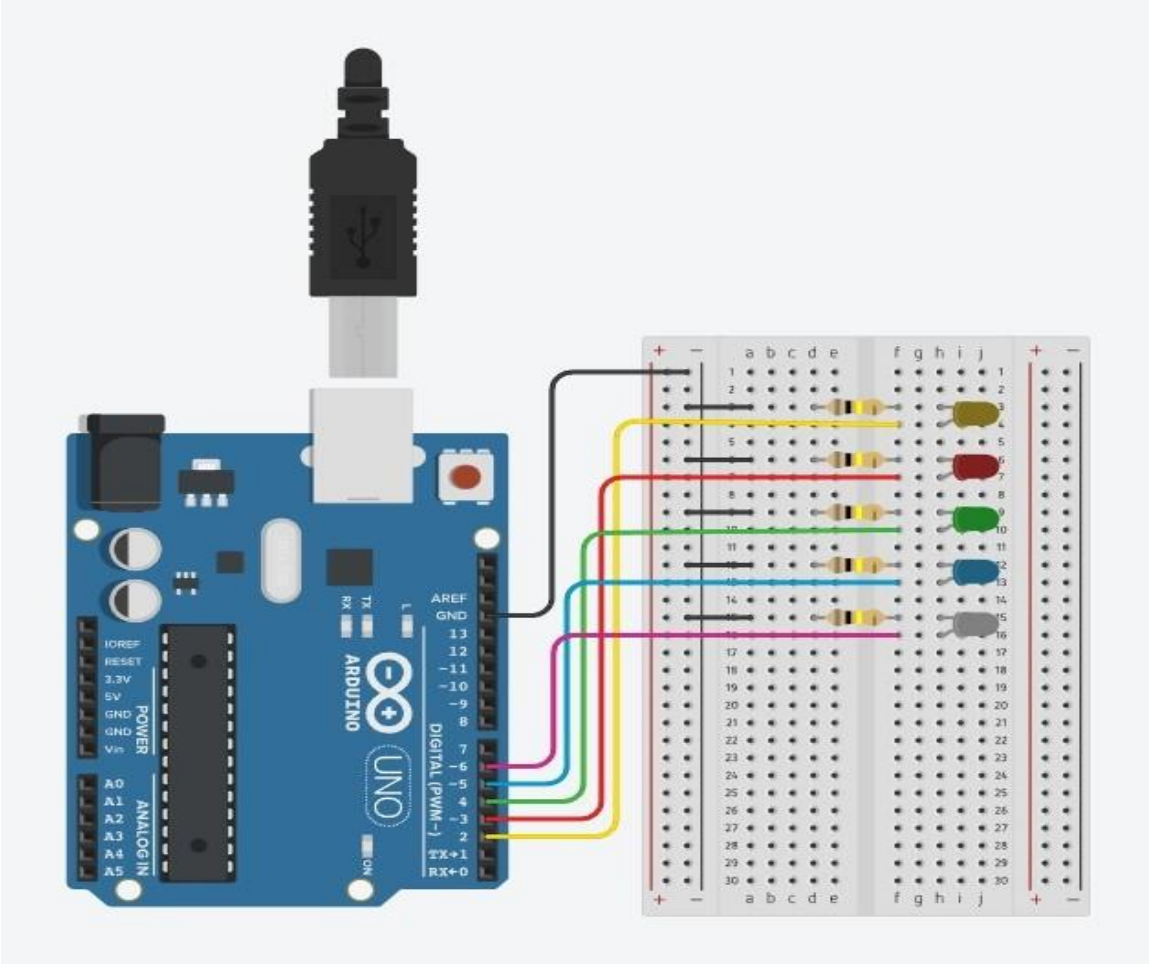
Experiment 1: Develop a program to blink 5 LEDs back and forth.

AIM: Develop a program to blink 5 LEDs back and forth.

COMPONENTS:

SL.NO	Name	Quantity
1	Arduino Uno	**1**
2	Jumper Cable	**6**
3	Bread Board	**1**
4	LED	**5**
5	Resistance (800 Ω)	**5**

CIRCUIT DIAGRAM:



SETUP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins are connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE, then go to Tools and select appropriate Arduino board.

d) Select Tools, then select the Port and choose the COM port to which the board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
void setup() {
pinMode(2, OUTPUT);
pinMode(3, OUTPUT);
pinMode(4, OUTPUT);
pinMode(5, OUTPUT);
pinMode(6, OUTPUT);
}
void loop() {
digitalWrite(2, HIGH);
delay(1000);
digitalWrite(2, LOW);
delay(100);

digitalWrite(3, HIGH);
delay(1000);
digitalWrite(3, LOW);
delay(100);
```

```
digitalWrite(4, HIGH);
delay(1000);
digitalWrite(4, LOW);
delay(100);

digitalWrite(5, HIGH);
delay(1000);
digitalWrite(5, LOW);
delay(100);

digitalWrite(6, HIGH);
delay(1000);
digitalWrite(6, LOW);
delay(100);

digitalWrite(5, HIGH);
delay(1000);
digitalWrite(5, LOW);
delay(100);

digitalWrite(4, HIGH);
delay(1000);
digitalWrite(4, LOW);
delay(100);

digitalWrite(3, HIGH);
delay(1000);
```

```
digitalWrite(3, LOW);
delay(100);

digitalWrite(2, HIGH);
delay(1000);
}
```

RESULT: Successfully demonstrated blink 5 LEDs.

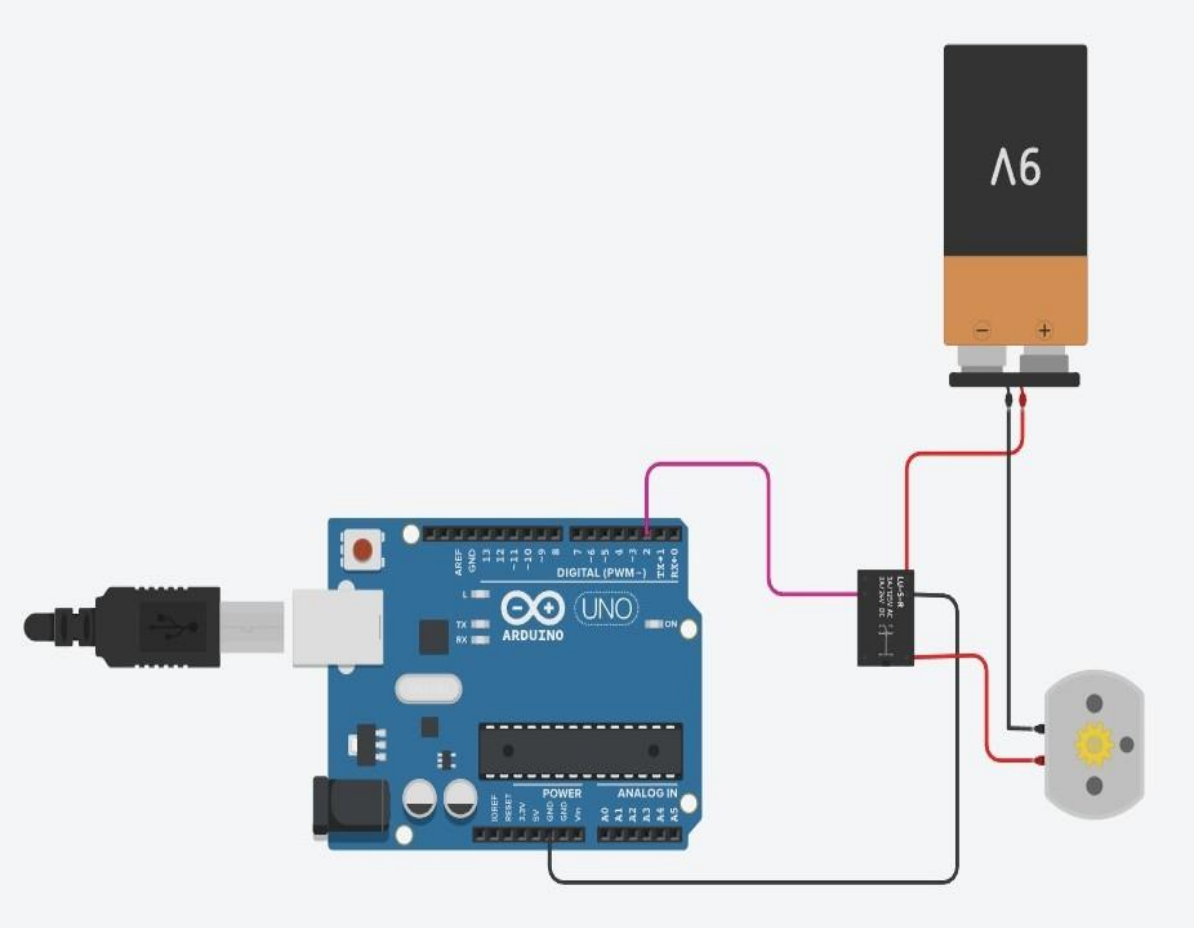
Experiment 2. Develop a program to interface a relay with Arduino board.

AIM: Develop a program to interface a relay with Arduino board.

COMPONENTS:

SL NO	Name	Quantity
1	Arduino Uno	**1**
2	Jumper Cable	**6**
3	Bread Board	**1**
4	DC Motor	**1**
5	Relay SPDT	**1**
6	9V Battery	**1**

CIRCUIT DIAGRAM:



SETUP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins are connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then go to Tools and select appropriate Arduino board.

d) Select Tools then select the Port and select the COM port to which the board is connected.

e) Type sketch (Program) and upload to board.

CODE:

```
void setup()
{
pinMode(2, OUTPUT);
}
void loop()
{
digitalWrite(2, HIGH);
delay(10000);
digitalWrite(2, LOW);
delay(5000);
}
```

RESULT: Successfully demonstrated interface a relay with Arduino board.

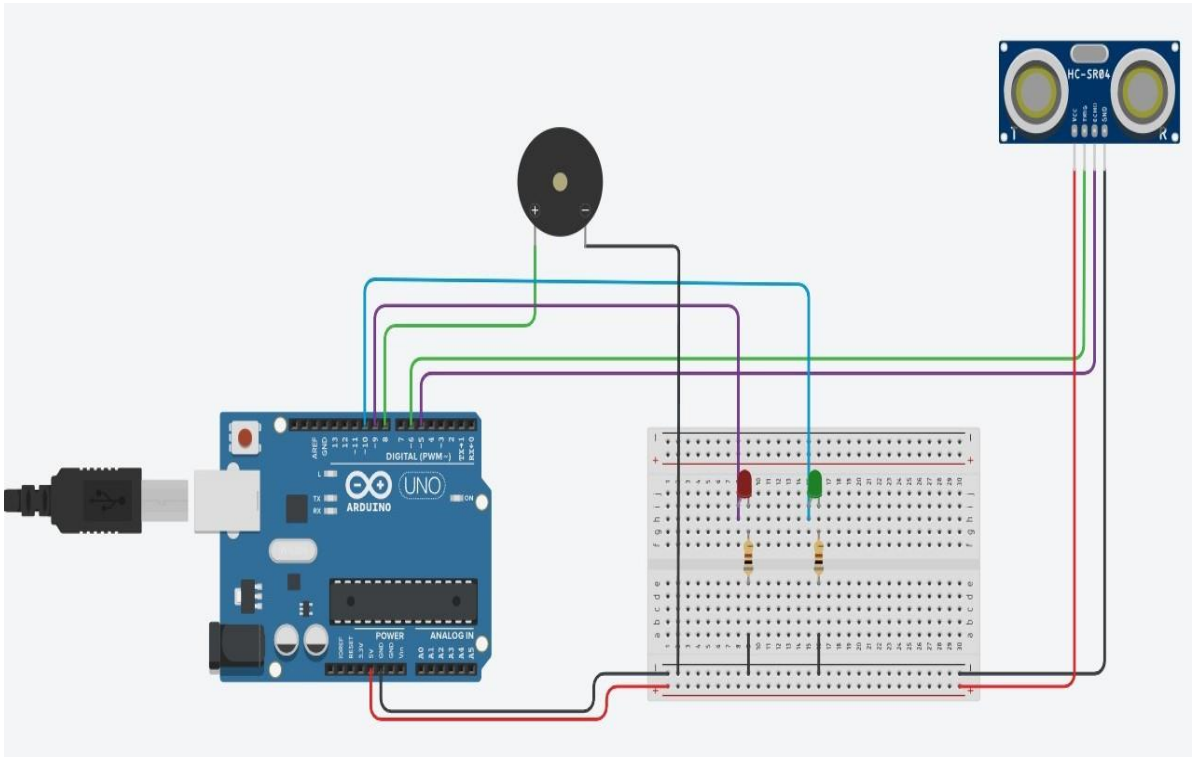
Experiment 3. Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

AIM: Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

COMPONENTS:

SL.NO	Name	Quantity
1	Arduino Uno	**1**
2	Jumper Cable	**6**
3	Bread Board	**1**
4	LED	**2**
5	Resistance (800 Ω)	**2**
6	Ultrasonic Distance Sensor (4-pin)	**1**
7	Piezo	**1**

CIRCUIT DIAGRAM:



SETUP:

a) Connect the circuit as per circuit.

b) Make sure VCC and Ground pins are connected properly to avoid any damage to Arduino board.

c) Open Arduino IDE then go to Tools and select appropriate Arduino board.

d) Select Tools then select the Port and select the COM port to which the board is connected.

e) Type sketch (Program) and upload to board.

CODE:

